

Table. Statistical significance (paired t -test) comparing MAGNET with baseline methods under 80% missingness ratio on simulated cancer data (corresponding to Figure 4). An asterisk (*) indicates $p < 0.05$.

Missing Pattern	p -value (vs M3Care)	p -value (vs MUSE)
Random Masking	0.0055*	0.0002*
Missing All Modalities	0.0230*	0.0011*
Intact mRNA, Missing DNA and Protein	0.0832	0.0287*
Intact Protein, Missing DNA and mRNA	0.3456	0.0133*
Intact DNA, Missing mRNA and Protein	0.0637	0.0855

Table. Statistical significance (paired t -test) comparing MAGNET with its ablated individual components across datasets (corresponding to Table 3). The overall p -value is computed using paired t -tests over all folds across datasets. An asterisk (*) indicates $p < 0.05$.

ID	Method	p -value (BRCA)	p -value (BLCA)	p -value (OV)	Overall p -value
A1	MAGNET w/o PMMHA	0.0161*	0.2457	0.2302	0.0528
A2	MAGNET w/o GNN	0.1543	0.2519	0.1890	0.0740
A3	MAGNET w/o Edge Feature	0.0249*	0.8372	0.4600	0.3266
A4	MAGNET w/o KL Loss	0.1087	0.3885	0.0993	0.1143
A5	MAGNET w/ GAT	0.0080*	0.1269	0.1492	0.0007*
A6	MAGNET w/ GCN	0.0012*	0.1153	0.1708	0.0013*
A7	MAGNET w/ GIN	0.0167*	0.0175*	0.1987	0.0003*